

# DW02\_mult\_2\_stage

## Two-Stage Pipelined Multiplier

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### Features and Benefits

- Parameterized word length
- Unsigned and signed (two's-complement) data operation
- Two-stage pipelined architecture
- Automatic pipeline retiming

### Revision History

### Description

DW02\_mult\_2\_stage is a two-stage pipelined multiplier.  
DW02\_mult\_2\_stage multiplies the operand A by B to produce a product (PRODUCT) with a latency of one clock (CLK) cycle.

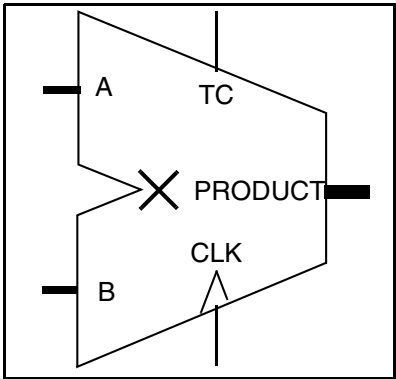


Table 1-1 Pin Description

| Pin Name | Width                                | Direction | Function                                                                                                     |
|----------|--------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| A        | <i>A_width</i> bits                  | Input     | Multiplier                                                                                                   |
| B        | <i>B_width</i> bits                  | Input     | Multiplicand                                                                                                 |
| TC       | 1 bit                                | Input     | Two's complement control <ul style="list-style-type: none"><li>■ 0 = Unsigned</li><li>■ 1 = Signed</li></ul> |
| CLK      | 1 bit                                | Input     | Clock                                                                                                        |
| PRODUCT  | <i>A_width</i> + <i>B_width</i> bits | Output    | Product ( $A \times B$ )                                                                                     |

Table 1-2 Parameter Description

| Parameter | Values   | Description      |
|-----------|----------|------------------|
| A_width   | $\geq 1$ | Word length of A |
| B_width   | $\geq 1$ | Word length of B |

Table 1-3      Synthesis Implementations

| Implementation Name | Function                                      | License Feature Required |
|---------------------|-----------------------------------------------|--------------------------|
| str                 | Area or delay optimized flexible architecture | DesignWare               |

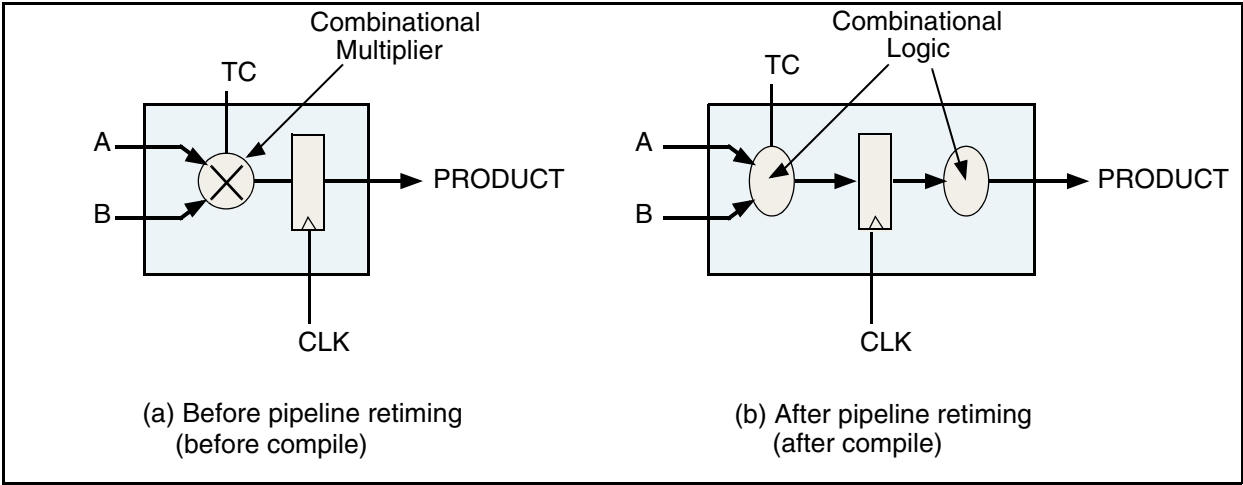
Table 1-4      Simulation Models

| Model                                 | Function                             |
|---------------------------------------|--------------------------------------|
| DW02.DW02_MULT_2_STAGE_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw02/src/DW02_mult_2_stage_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW02_mult_2_stage.v        | Verilog simulation model source code |

The control signal, TC, determines whether the input and output data is interpreted as unsigned (TC = 0) or signed (TC = 1) numbers.

Automatic pipeline retiming ensures optimal placement of pipeline registers within the multiplier to achieve maximum throughput.

Figure 1-1      DW02\_mult\_2\_stage Block Diagram



Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW02_mult_2_stage_inst is
  generic ( inst_A_width : POSITIVE := 8;
            inst_B_width : POSITIVE := 8 );
  port ( inst_A  : in std_logic_vector(inst_A_width-1 downto 0);
        inst_B  : in std_logic_vector(inst_B_width-1 downto 0);
        inst_TC  : in std_logic;
        inst_CLK : in std_logic;
        PRODUCT_inst : out std_logic_vector(inst_A_width+inst_B_width-1 downto 0)
        );
end DW02_mult_2_stage_inst;

architecture inst of DW02_mult_2_stage_inst is
begin

  -- Instance of DW02_mult_2_stage
  U1 : DW02_mult_2_stage
    generic map ( A_width => inst_A_width,  B_width => inst_B_width )
    port map ( A => inst_A,  B => inst_B,  TC => inst_TC,
              CLK => inst_CLK,  PRODUCT => PRODUCT_inst );
end inst;

-- pragma translate_off
configuration DW02_mult_2_stage_inst_cfg_inst of DW02_mult_2_stage_inst is
  for inst
  end for; -- inst
end DW02_mult_2_stage_inst_cfg_inst;
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW02_mult_2_stage_inst( inst_A, inst_B, inst_TC,
                              inst_CLK, PRODUCT_inst );

    parameter A_width = 8;
    parameter B_width = 8;

    input [A_width-1 : 0] inst_A;
    input [B_width-1 : 0] inst_B;
    input inst_TC;
    input inst_CLK;
    output [A_width+B_width-1 : 0] PRODUCT_inst;

    // Instance of DW02_mult_2_stage
    DW02_mult_2_stage #(A_width, B_width)
        U1 ( .A(inst_A),    .B(inst_B),    .TC(inst_TC),
            .CLK(inst_CLK),    .PRODUCT(PRODUCT_inst) );

endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                        |
|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | ■ Updated date and version; no technical changes                                                                                                                               |
| December 2023  | DWBB_202312.0 | ■ Updated date and version; no technical changes                                                                                                                               |
| July 2020      | DWBB_201912.5 | ■ Removed the “Disabling Clock Monitor Messages” section                                                                                                                       |
| October 2019   | DWBB_201903.5 | ■ Updated description of ‘str’ implementation in <a href="#">Table 1-3</a> on page 2<br>■ Added the “Disabling Clock Monitor Messages” section                                 |
| January 2019   | DWBB_201806.5 | ■ Updated example in “ <a href="#">HDL Usage Through Component Instantiation - VHDL</a> ” on page 3<br>■ Added this Revision History table and the document links on this page |

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